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Diagnostics and troubleshooting

These controllers detect a wide variety of faults or error conditions. Faults can be detected by the DS14 PC programmer or DS13 handheld programmer.

8.1 DIAGNOSTICS

Diagnostics information can be obtained in the three ways: (1) by listening the buzzer sounds from controller, (2) by reading the display on a DS13 handheld programmer, or (3) by reading the display on DS14 PC programmer.

The buzzer if is enabled will buzz to alert any fault occurs. The long and short buzzs provides user a fault code to help them identify and resolve fault. Note: the buzzer only indicates the latest fault. When there are multiple faults occurred, the next buzzer can be triggered only after previous buzzer is cleared.

The DS13 handheld programmer or DS14 PC programmer will display all faults that are currently set as well as a history of the faults that have been set since the history log was last cleared. The programmer displays the faults by name.

8.2 TROUBLESHOOTING

The troubleshooting chart, Table 5, provides the following information on all the controller faults:

- (1) fault code
- (2) buzzs
- (3) fault name
- (4) troubleshoot
- (5) possible cause
- (6) fault set / clear conditions

Whenever a fault is encountered and no wiring or vehicle fault can be found, shut off KSI and turn it back on to see if the fault clears. If it does not, shut off KSI and remove the connector. Check the connector for corrosion or damage, clean it if necessary, and re-insert it.

COD E	BUZZS	Fault Name	TROUBLESHOOT	POSSIBLE CAUSE	SET/CLEAR CONDITIONS

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0x12	1 long 3 short	Controller Overcurrent	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1. External short of phase U,V, or W motor connections; 2. Motor parameters are mis-tuned; 3. Controller defective	<i>Set:</i> Phase current exceeded the current measurement limit. <i>Clear:</i> Cycle KSI
0x13	1 long 6 short	Current sensor fault 1	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1. Controller defective, 2. Leakage to vehicle frame from phase U, V, or W (short in motor stator)	<i>Set:</i> Controller current sensors have invalid offset reading. <i>Clear:</i> Cycle KSI
0x13	1 long 6 short	Current sensor fault 2	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1. Controller defective, 2. Leakage to vehicle frame from phase U, V, or W (short in motor stator)	<i>Set:</i> Controller current sensors have invalid offset reading. <i>Clear:</i> Cycle KSI
0x14	1 long 2 short	Precharge Failed	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1.Precharge circuit failed 2.External load on capacitor bank	<i>Set:</i> Precharge failed to charge the capacitor bank to the KSI voltage. <i>Clear:</i> Cycle Interlock input or KSI
0x15	3 long 1 short	Controller Severe Undertemp	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1. Controller is operating in an extreme temperature environment. 2.Controller temp sensor fault.	<i>Set:</i> Temperature below -40°C. <i>Clear:</i> Bring heatsink temperature above -40°C, and cycle interlock or KSI

0x16	1 long 4 short	Controller Severe Overtemp	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1. Controller is operating in an extreme temperature environment. 2. Controller temp sensor fault. 3. Excessive load on vehicle ; 4. Improper mounting of controller	Set: Heatsink temperature above +90°C. Clear: Bring heatsink temperature below +90°C, and cycle interlock or KSI
0x17	1 long 9 short	Severe B+ Undervoltage	Reduced drive torque.	1. Battery Menu parameters are misadjusted, 2. Non-controller system drain on battery, 3. Battery resistance too high, 4. Battery disconnected while driving, 5. Blown B+ fuse or main contactor did not close.	Set: Capacitor bank voltage dropped below the Severe Undervoltage limit with FET bridge enabled. Clear: Bring capacitor voltage above Severe Undervoltage limit.
0x18	1 long 10 short	Severe B+ Overvoltage	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1. Battery Menu parameters are misadjusted, 2. Battery resistance too high 3. Battery disconnected while regen braking.	Set: Capacitor bank voltage exceeded the Severe Overvoltage limit with FET bridge enabled. Clear: Bring capacitor voltage below Severe Overvoltage limit, and cycle KSI.
0x19	2 long 7 short	Motor Temp Sensor Fault	MaxSpeed reduced (LOS, Limited Operating Strategy), and motor temperature cutback disabled.	Controller internal fault	Set: Detected controller internal fault, Clear: cycle KSI
0x21	2 long 11 short	Controller Undertemp Cutback	Reduced drive and brake torque.	1. Controller is operating in an extreme temperature environment. 2. Controller temp sensor fault.	Set: Temperature below -30°C. Clear: Bring heatsink temperature above -30°C
0x22	2 long 12 short	Controller Overtemp Cutback	Reduced drive and brake torque.	1. Controller is performance-limited at this temperature. 2. Controller is operating in an extreme temperature environment. 3. Controller temp sensor fault, 4. Excessive load on vehicle, 5. Improper mounting of controller.	Set: Heatsink temperature exceeded 80°C, Clear: Bring heatsink temperature below 80°C.

0x23	2 long 14 short	B+ Undervoltage Cutback	Reduced drive torque.	1. Normal operation. Fault shows that the batteries need recharging. Controller is performance limited at this voltage. 2. Battery parameters are misadjusted. 3. Non-controller system drain on battery. 4. Battery resistance too high, 5. Battery disconnected while driving 6. Fuse or main contactor did not close.	<i>Set:</i> Capacitor bank voltage dropped below the Undervoltage limit. <i>Clear:</i> Bring capacitor voltage above the Undervoltage limit.
0x24	2 long 13 short	B+ Overvoltage Cutback	Reduced brake torque.	1. Normal operation. Fault shows that regen braking currents elevated the battery voltage during regen braking. Controller is performance limited at this voltage. 2. Battery parameters are misadjusted. 3. Battery resistance too high for given regen current. 4. Battery disconnected while regen braking.	<i>Set:</i> Capacitor bank voltage exceeded the Overvoltage limit. <i>Clear:</i> Bring capacitor voltage below the Overvoltage limit.
0x25	3 long 3 short	+5V Supply Failure		1. External load impedance on the +5V supply is too low. 2. The 5V supply module of controller is defective.	<i>Set:</i> Capacitor bank voltage out of the range +5V$\pm$10%. <i>Clear:</i> Bring capacitor voltage within range.
0x28	1 long 11 short	Motor Temp Hot Cutback	Reduced drive torque.	1. Motor temperature is at or above the programmed Temperature Hot setting, and the current is being cut back. 2. Motor Temperature Control Menu parameters are mis-tuned. 3. If the application doesn't use a motor thermistor, Temp Compensation and Temp Cutback should be programmed Off.	<i>Set:</i> Motor temperature is at or above the Temperature Hot parameter setting. <i>Clear:</i> Bring the motor temperature within range.
0x29	1 long 15 short	Motor Temp Sensor Fault	MaxSpeed reduced (LOS, Limited Operating Strategy), and motor temperature cutback disabled.	1. Motor thermistor is not connected properly, 2. Doesn't use a thermistor, and Motor Temp Sensor Enable is not programmed off.	<i>Set:</i> Motor thermistor input is at the voltage rail (0 or 10V). <i>Clear:</i> Bring the motor thermistor input voltage within range.
0x31	1 long 5 short	Coil1 Driver Open/Short	ShutdownDriver 1	1. Open or short on driver load 2. Dirty connector pins, 3. Bad crimps or faulty wiring.	<i>Set:</i> Driver 1 is either open or shorted. <i>Clear:</i> Correct open or short, and cycle driver

0x31	1 long 5 short	Main Open/Short	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1.Open or short on driver load 2.Dirty connector pins, 3.Bad crimps or faulty wiring.	Set: Driver 1 is either open or shorted. <i>Clear:</i> Correct open or short, and cycle driver
0x32	3 long 6 short	Coil2 Driver Open/Short	ShutdownDriver 2	1.Open or short on driver load 2.Dirty connector pins, 3.Bad crimps or faulty wiring.	<i>Set:</i> Driver 2 (pin 5) is either open or shorted. This fault can be set only when EM Brake Type = 0. <i>Clear:</i> Correct open or short, and cycle driver.
0x32	3 long 6 short	EMBrake Open/Short	Shutdown EM Brake, Shutdown Throttle, Full Brake,	1.Open or short on driver load 2.Dirty connector pins, 3.Bad crimps or faulty wiring.	<i>Set:</i> Electromagnetic brake driver is either open or shorted. This fault can be set only when EM Brake Type > 0. <i>Clear:</i> Correct open or short, and cycle driver.
0x36	1 long 12 short	Encoder Fault	ShutdownEMBrake; ShutdownThrottle.	1.Motor encoder failur; 2.Bad crimps or faulty wiring.	Set: Motor encoder phase failure detected, <i>Clear:</i> Cycle KSI.
0x37	1 long 14 short	Motor Open	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1.Motor phase is open, 2.Bad crimps or faulty wiring.	<i>Set:</i> Motor phase U, V, or W detected open. <i>Clear:</i> Cycle KSI.
0x39	4 long 2 short	Main Contactor Did Not Close	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1.Main contactor did not close; 2.Main contactor tips are oxidized, burned, or not making good contact. 3.External load on capacitor bank (B+ connection terminal) that prevents capacitor bank from charging. 4.Blown B+ fuse.	<i>Set:</i> With the main contactor commanded closed, the capacitor bank voltage (B+ connection terminal) did not charge to B+. <i>Clear:</i> Cycle KSI.
0x41	4 long 3 short	Throttle Wiper High	Shutdown Throttle	1.Throttle pot wiper voltage too high, 2. Improper throttle parameter setting.	Set: Throttle pot wiper voltage is higher than the high fault threshold, <i>Clear:</i> Bring throttle pot wiper voltage below the fault threshold.

0x42	4 long 4 short	Throttle Wiper Low	Shutdown Throttle	1. Throttle pot wiper voltage too low, 2. Improper throttle parameter setting.	Set: Throttle pot wiper voltage is lower than the high fault threshold, Clear: Bring throttle pot wiper voltage above the fault threshold.
0x43	4 long 5 short	Pot2 Wiper High	FullBrake	Pot2 wiper voltage too high	Set: Pot2 wiper voltage is higher than the high fault threshold. Clear: Bring Pot2 wiper voltage below the fault threshold.
0x44	4 long 6 short	Pot2 Wiper Low	FullBrake	Pot2 wiper voltage too low	Set: Pot2 wiper voltage is lower than the high fault threshold. Clear: Bring Pot2 wiper voltage above the fault threshold.
0x46	2 long 3 short	EEPROM Failure	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump, Shutdown Interlock, ShutdownDriver1, ShutdownDriver2, ShutdownDriver3, ShutdownDriver4, ShutdownPD;	Failure to write to EEPROM memory.	<i>Set:</i> Controller operating system failed to write to EEPROM memory. <i>Clear:</i> Upgrade the correct software and matching parameter default settings into the controller and cycle KSI.
0x47	long	HPD/Sequencing Fault	Shutdown Throttle	1. KSI, interlock, direction, and throttle inputs applied in incorrect sequence. 2. Faulty wiring, crimps, or switches at KSI, interlock, direction, or throttle inputs.	<i>Set:</i> HPD (High Pedal Disable) or sequencing fault caused by incorrect sequence of KSI, interlock, direction, and throttle inputs. <i>Clear:</i> Reapply inputs in correct sequence.
0x47	long	Emer Rev HPD	Shutdown EM Brake, Shutdown Throttle	Emergency Reverse operation has concluded, but the throttle, forward and reverse inputs, and interlock have not been returned to neutral.	<i>Set:</i> At the conclusion of Emergency Reverse, the fault was set because various inputs were not returned to neutral. <i>Clear:</i> Clear the interlock, throttle, and direction inputs.

0x48	4 long 8 short	Following Error Fault	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump, Shutdown Interlock, ShutdownDriver1, ShutdownDriver2, ShutdownDriver3, ShutdownDriver4, ShutdownPD;	1. The Following Error Limit has been exceeded for the Following Error Time. 2. Incorrect or overly restrictive Following Error Limit and Following Error Time parameter settings, 3. Motor or drivetrain rotation obstruction or degradation.	<i>Set:</i> The Following Error Limit has been exceeded for the Following Error Time. <i>Clear:</i> Cycle KSI.
0x49	4 long 9 short	Parameter Change Fault	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	This is a safety fault caused by a change in certain parameter settings so that the vehicle will not operate until KSI is cycled. For example, if a user changes the Throttle Type this fault will appear and require cycling KSI before the vehicle can operate.	<i>Set:</i> Adjustment of a parameter setting that requires cycling of KSI. <i>Clear:</i> Cycle KSI.
0x72	7 long 2 short	CAN Fault	ShutdownInterlock	Time between CAN PDO messages received exceeded the PDO Timeout Period.	<i>Set:</i> Time between CAN PDO messages received exceeded the PDO Timeout Period. <i>Clear:</i> Cycle KSI or receive CAN NMT message.
0x73	7 long 3 short	Stall Detected Fault	Shutdown EM Brake Shutdown Throttle	1. Poor wiring or wrong wiring. 2. The encoder line sequence is wrong. 3. The number of encoder pulses is wrong. 4. Mechanical failure. 5. The motor is wired incorrectly. 6. The motor load is too large.	<i>Set:</i> the motor current is too high and the motor speed is too low. <i>Clear:</i> check the cause of the motor stall and restart the key switch.
0x76	2 long 9 short	+12V Supply Fault		1. External load on the +12V supplies draws either too much or too little current. 2. The controller +12V power supply module defective.	<i>Set:</i> Voltage out of the range +12V±10%. <i>Clear:</i> Bring voltage within range.

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0x77	1 long 13 short	Throttle Switch Fault	ShutdownThrottle	1.Accelerate Switch no output, 2.Poor or incorrect accelerate switch wiring, 3.Incorrect parameter setting.	<i>Set:</i> No throttle switch detected when Throttle Switch Enable=on, and throttle >25%. <i>Clear:</i> Cycle KSI.
0x78	1 long 8 short	BMS Fault	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	BMS fault.	<i>Set:</i> Detected BMS fault. <i>Clear:</i> Cycle KSI.
0x81	3 long 2 short	KSI Switch Voltage Error	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1. Bad crimps or faulty wiring; 2. Non-controller system drain on battery 3.Low power circuit resistance too high. 4. Cut low power circuit power off when drive.	<i>Set:</i> KSI voltage exceeded range. <i>Clear:</i> Cycle KSI.
0x82	2 long 6 short	Controller Fault 4	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	Controller internal fault	<i>Set:</i> Detected controller internal fault. <i>Clear:</i> Cycle KSI.
0x83	2 long 2 short	Controller Fault 1	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	Controller internal fault	<i>Set:</i> Detected controller internal fault. <i>Clear:</i> Cycle KSI.
0x83	2 long 2 short	Controller Fault 2	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	Controller internal fault	<i>Set:</i> Detected controller internal fault. <i>Clear:</i> Cycle KSI.

0x83	2 long 2 short	Controller Fault 3	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	Controller internal fault	Set: Detected controller internal fault. Clear: Cycle KSI.
0x84	2 long 1 short	Controller Fault 7		Controller internal fault	Set: Detected controller internal fault. Clear: Cycle KSI.
0x84	2 long 1 short	Controller Fault 8		Controller internal fault	Set: Detected controller internal fault. Clear: Cycle KSI.
0x84	2 long 1 short	Controller Fault 9		Controller internal fault	Set: Detected controller internal fault. Clear: Cycle KSI.
0x85	2 long 4 short	Controller Fault 10		Controller internal fault	Set: Detected controller internal fault. Clear: Cycle KSI.
0x85	2 long 4 short	Controller Fault 11		Controller internal fault	Set: Detected controller internal fault. Clear: Cycle KSI.
0x86	2 long 5 short	Controller Fault 5	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	Controller internal fault	Set: Detected controller internal fault. Clear: Cycle KSI.
0x86	2 long 10 short	Controller Fault 6	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	Controller internal fault	Set: Detected controller internal fault. Clear: Cycle KSI.
0x87	2 long 8 short	Controller Fault 12		Controller internal fault	Set: Detected controller internal fault. Clear: Cycle KSI.

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0x88	8 long 8 short	Motor Characterization Fault	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1. Motor characterization failed during characterization process. 0= none, 1= encoder signal seen, but step size not determined; set Encoder Step Size manually. 2= motor temp sensor fault. 3= motor temp hot cutback fault. 4= controller overtemp cutback fault. 5= controller undertemp cutback fault. 6= undervoltage cutback fault 7= severe overvoltage fault 8= encoder signal not seen, one or both channels missing, 9= motor parameters out of characterization range.	Set: Motor characterization failed during the motor characterization process. Clear: Correct fault; cycle KSI.
0x89	8 long 9 short	Motor Type Fault	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	The Motor_Type parameter value is out of range.	Set: Motor_Type parameter is set to an illegal value. Clear: Set Motor_Type to correct value and cycle KSI.
0x91	9 long 1 short	Encoder Pulse Count Fault	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump Shutdown Interlock, ShutdownDriver1, ShutdownDriver2, ShutdownDriver3, ShutdownDriver4, ShutdownPD;	Encoder Steps parameter does not match the actual motor encoder.	Set: Motor lost IFO control and accelerated without throttle command. Clear: Ensure the Encoder Steps parameter matches the actual encoder; cycle KSI.

0x92	9 long 2short	EM Brake Failed to Set	Shutdown EM Brake, Shutdown Throttle,	1. Vehicle movement sensed after the EM Brake has been commanded to set. 2. EM Brake will not hold the motor from rotating.	Set: After the EM Brake was commanded to set and time has elapsed to allow the brake to fully engage, vehicle movement has been sensed. Clear: Activate the throttle.
0x94	9 long 4 short	EMR Rev Timeout	Shutdown EM Brake, Shutdown Throttle,	1. Emergency Reverse was activated and concluded because the EMR Timeout timer has expired. 2. The emergency reverse input is stuck On.	Set: Emergency Reverse was activated and ran until the EMR Timeout timer expired. Clear: Turn the emergency reverse input Off.
0x98	9 long 8 short	Illegal Model Number	Shutdown Motor, Shutdown main contactor, Shutdown EM Brake, Shutdown Throttle, Full Brake, Shutdown Pump	1. Model_Number variable contains illegal value, 2. Software and hardware do not match. 3. Controller defective.	Set: Illegal Model_Number variable; when KSI cycles, a check is made to confirm a legal Model_Number, and a fault is issued if one is not found. Clear: Upgrade appropriate software for your controller model.